MainCrafts Technology – AI & ML Internship

Task 1: Build & Evaluate a Linear Regression Model (House Price Predictor)

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1. Objective

The objective is to implement a complete machine-learning workflow for house-price prediction using the California Housing dataset. The workflow covers data loading, exploratory data analysis, preprocessing, train/test splitting, model training, evaluation, visualization, and reporting.

2. Dataset

The California Housing dataset contains 20,640 observations, eight numeric predictive features, and the target variable **MedHouseVal**. The target represents median house value in units of $100,000. The features are MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, and Longitude.

3. Exploratory Data Analysis

- Checked dataset dimensions, data types, summary statistics, missing values, and duplicate rows.
- Plotted distributions of the numeric variables.
- Created a correlation heatmap to understand relationships between variables.
- Examined the target distribution and the relationship between median income and median house value.
- No categorical encoding is required because the standard dataset contains numeric features.

4. Preprocessing and Split

The target column MedHouseVal was separated from the eight input features. The data was split into 80% training data and 20% test data using **random_state=42** so that the experiment is reproducible.

5. Model Development

A scikit-learn **LinearRegression** model was fitted using the training set. The trained model was then used to predict median house values for the unseen test set.

6. Evaluation Metrics

Metric	Reference result	Interpretation
MAE	0.5332	Average absolute prediction error; lower is better.
RMSE	0.7456	Penalizes larger errors; lower is better.
R^2	0.5758	About 57.58% of target variance is explained by the model.

The reference results above correspond to the standard California Housing dataset with an 80/20 train-test split and random_state=42. The notebook calculates the metrics directly; the final submitted report should match the values printed by the executed notebook.

7. Visualization

The notebook contains an Actual-vs-Predicted scatter plot and a residual plot. A good regression model generally places predictions near the diagonal line in the Actual-vs-Predicted chart and keeps residuals relatively centered around zero without a strong systematic pattern.

8. Model Saving

The notebook also includes an optional step to save the trained Linear Regression model using joblib. This can be reused later for a small prediction interface.

9. Findings

- Linear Regression provides a clear and reproducible baseline for California house-value prediction.
- Median income is an important predictor of house value, while geographic variables also contribute useful information.
- The R^2 reference value of 0.5758 indicates that the linear model explains a substantial portion, but not all, of the variation in house values.
- The remaining error suggests that house prices contain nonlinear relationships and interactions that a simple linear model cannot fully capture.

10. Improvement Ideas

- Compare Linear Regression with Ridge and Lasso regression.
- Use cross-validation to obtain a more robust estimate of generalization performance.
- Try nonlinear models such as Decision Tree, Random Forest, or Gradient Boosting.
- Perform careful feature engineering and outlier analysis.
- Build a small Streamlit UI for entering feature values and obtaining a prediction.

11. Conclusion

This project demonstrates the end-to-end machine-learning lifecycle requested in Task 1: loading and exploring data, preparing features, splitting data, training a Linear Regression model, evaluating it with MAE/RMSE/R^2, visualizing predictions and residuals, and saving the model. The project is suitable as a baseline portfolio project and can be extended with stronger regression algorithms.

Source note: Task requirements are based on the MainCrafts Technology internship task guide supplied with this project. Dataset definitions and reference metric values were cross-checked against scikit-learn documentation and public implementations.